

Test and Reliability Exercise Sheet 3

Exercise 1

Consider the small circuit in Figure 1. Your task is to perform a *fault* parallel simulation of the (three-valued) input vector $(a, b) = (1, X)$ for the following stuck-at-faults: s-a-0 and s-a-1 on line a , s-a-1 on line f , and s-a-0 on line g . Use a word size of 16.

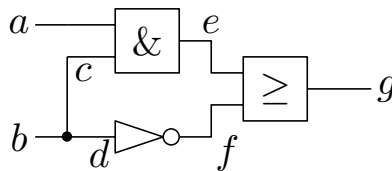


Figure 1: Small circuit

- Give the masks I_a, S_a, I_f, S_f, I_g , and S_g .
- Perform the simulation: For each line state the word and use two bits for each value.
- State the values at the output g in the fault free and the faulty cases. Which faults can be detected?
- Now, simulate the circuit again. This time use both possible assignments for the input vector $(a, b) = (1, X)$, i.e. $(a, b) = (1, 0)$ and $(a, b) = (1, 1)$. Which faults can be detected? Discuss the results.

Exercise 2

In the lecture the D-algorithm was presented (see also slides 3:57ff), where first the D-frontier is propagated, followed by the justification of the J-frontier. Consider two modified versions of the D-algorithm:

- 1.) First the J-frontier is justified and afterwards the D-frontier is propagated.
 - 2.) D- and J-frontier are propagated/justified in parallel.
- a) Discuss the difference of all these versions. Are the modified algorithms still complete? Is there a single version which is in general faster than the other ones? If yes: which one? If no: why?

Now consider again the circuit from exercise 2 of the first exercise sheet as shown in Fig. 2.

- b) How are controllability and observability measures calculated for OR- and NAND-gates (cf. the calculation for the AND-gate on slides 3:68ff)?
- c) Calculate the controllability and observability measures $C0$, $C1$ and O for every gate of the circuit in Fig. 2.
- d) Use the original D-algorithm and 5 valued logic in order to find a test pattern for the stuck-at-1-fault at gate G1 of the circuit in Fig. 2. If there is more than one gate in the D-frontier, use the observability measures from c) in order to decide the best option. Therefore, you should always choose the gate with the lowest observability measure. If two or more options have the same value, choose an arbitrary one. For each step, state the current D-frontier and J-frontier as well as the decisions and implications you have made. Compare your proceeding and result to exercise 2 of the first exercise sheet.
- e) Is it useful to utilize controllability and observability measures in order to find a test pattern for slow-to-fall transition delay faults? Briefly explain your solution.
- f) Apply PODEM in order to generate a test pattern for the same stuck-at-1-fault at gate G1. Which algorithm would you choose to solve the ATPG task at hand? D-Algorithm or PODEM? Explain your answer.

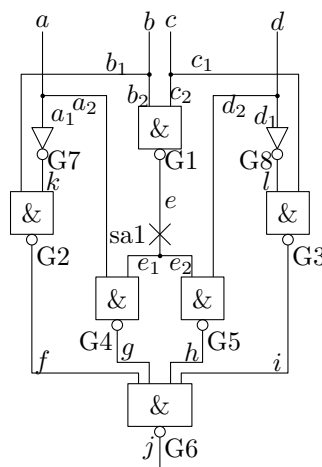


Figure 2: Faulty circuit

Exercise 3

- a) Assume the test set:

$$t_1 = 000x$$

$$t_2 = 101x$$

$$t_3 = 1x1x$$

$$t_4 = 1xx1$$

$$t_5 = x0x0$$

$$t_6 = x1x1$$

$$t_7 = xx01$$

Draw the compatibility graph for this test set. State the statically compacted test set with a minimal number of test patterns.

- b) Prove formally: All nodes of one clique in the compatibility graph can be compacted to one test pattern. In other words: For each clique $t_1, \dots, t_n$ there exists a test pattern $\tilde{t}$, which is compatible with t_i for each $i \in \{1, \dots, n\}$.

Hint: Use complete induction over the nodes in the clique.